

An implementation of Open-RMF in an industrial environment

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Part I

Introduction

This work presents a study on the new system, OPEN-RMF, which is still under active development but has reached a mature enough state to be deployed with robots at the Elettra Sincrotrone Trieste facility.

Chapter 1

Background and motivation

Elettra Sincontrone Trieste¹ is a major research facility that primarily provides users access to two light sources: the synchrotron Elettra and the free electron laser Fermi. In addition to operating these sources, the facility actively researches new technologies and techniques to enhance its capabilities and operations. Many recent advancements are based on robotics, which, especially after the COVID-19 pandemic and with the rapid rise of large language models (LLMs), have enabled new ways for users to interact with the infrastructure, including robot-assisted information retrieval and package delivery services. Previously, the facility developed a global planner, D-Star (D-star based optimized trajectory planner)[18], within the ROS 1 framework to enable robot navigation across large-scale environments. However, as ROS 1 has reached end-of-life, the codebase must now be migrated to ROS 2. Additionally, the facility employs robots from multiple vendors, each specialized in different tasks and not designed to coordinate with one another. In an environment that shares resources such as narrow passages and lifts, this lack of interoperability creates significant coordination challenges. To address this issue, Open-RMF² has been developed in recent years as a multiplatform system for controlling and managing heterogeneous fleets of robots.

¹<https://www.elettra.eu/>

²<https://www.open-rmf.org/>

Chapter 2

Research questions and objectives

The objectives of this thesis are as follows: to integrate a Rover Mini robot from the company RoverRobotics ¹ into the ROS 2 Nav2 ecosystem, to port the D-Star global planner from ROS 1 to ROS 2,



Figure 2.1: RoverRobotics Rover Mini

incorporate the robot into the Open-RMF framework for centralized fleet management and evaluate the suitability of the Open-RMF ecosystem for deployment at the Elettra Sincrotrone Trieste facility.

¹<https://roverrobotics.com/>

Chapter 3

Scope and limitations

The principal limitation is the current lack of access to lift control systems, which prevents direct integration and operation by the robot fleet. Additionally, there is no automated system for opening and closing doors. While Open-RMF offers a high degree of customizability, and human-in-the-loop interactions can be configured—such as triggering a light or sending a notification when a robot approaches a door—this method requires a human to manually open the door or operate the lift. However, such an approach is not ideal for achieving full autonomy.

Part II

Literature Review

In this section, we provide a description of the key technologies used in this thesis. We begin by describing the Robot Operating System (ROS) and its functionality, followed by an overview of the navigation system employed, and a discussion of how the Open-RMF fleet management system operates.

Chapter 4

Overview of ROS1 and ROS2

The Robotics Operating System (ROS)¹ is an open-source collection of tools and libraries designed to support the development of robotics applications. Despite its name, ROS is not an operating system (OS) in the traditional sense, but rather a software development kit (SDK) that provides developers with a comprehensive set of tools for building robotic systems. It offers several features typically associated with an OS, including hardware abstraction, low-level device control, implementation of commonly used functionalities, inter-process communication, and package management. The ROS ecosystem is built upon the following core components:

- **Middleware communication layer (RMW):** Built on the Data Distribution Service (DDS) standard, this layer enables nodes to exchange information using an anonymous publish/subscribe pattern. This design promotes fault isolation, separation of concerns, and modular interfaces. ROS also supports synchronous, Remote Procedure Call (RPC)-style communication between services.

While the default DDS implementation is eProsima Fast DDS (`rmw_fastrtps_cpp`), this work uses Eclipse Cyclone DDS (`rmw_cyclonedds_cpp`) due to its better compatibility with the Nav2 navigation stack in ROS Humble.

- **Development tools:** ROS provides a suite of tools for managing and enhancing software development, including utilities for launching, introspection, debugging, visualization, plotting, logging, and playback.
- **Multi-machine support** ROS includes tools and libraries for retrieving, building, writing, and executing code across multiple computers and OS.
- **Community and governance:** ROS is supported by an active and extensive global community, coordinated primarily by Open Robotics², which serves as the central organization for its development and maintenance.

¹<https://www.ros.org/>

²<https://www.openrobotics.org/>

4.1 Structure of ROS

The official ROS documentation³ provides comprehensive information about the system. All communication in ROS is handled through the **ROS Graph**, a peer-to-peer network of processes (called nodes) that execute concurrently and exchange information with one another. Some components of this graph are described below. Note that some of the functionalities described are specific to ROS 2 and are not available in ROS 1:

- **Nodes:** A node is a modular process that performs a specific task. Nodes communicate with each other through topics, services, actions, and parameters. In ROS 2, nodes can be implemented as **LifecycleNode**, which provide a managed state machine with defined transitions for startup and shutdown, enabling more deterministic behavior.

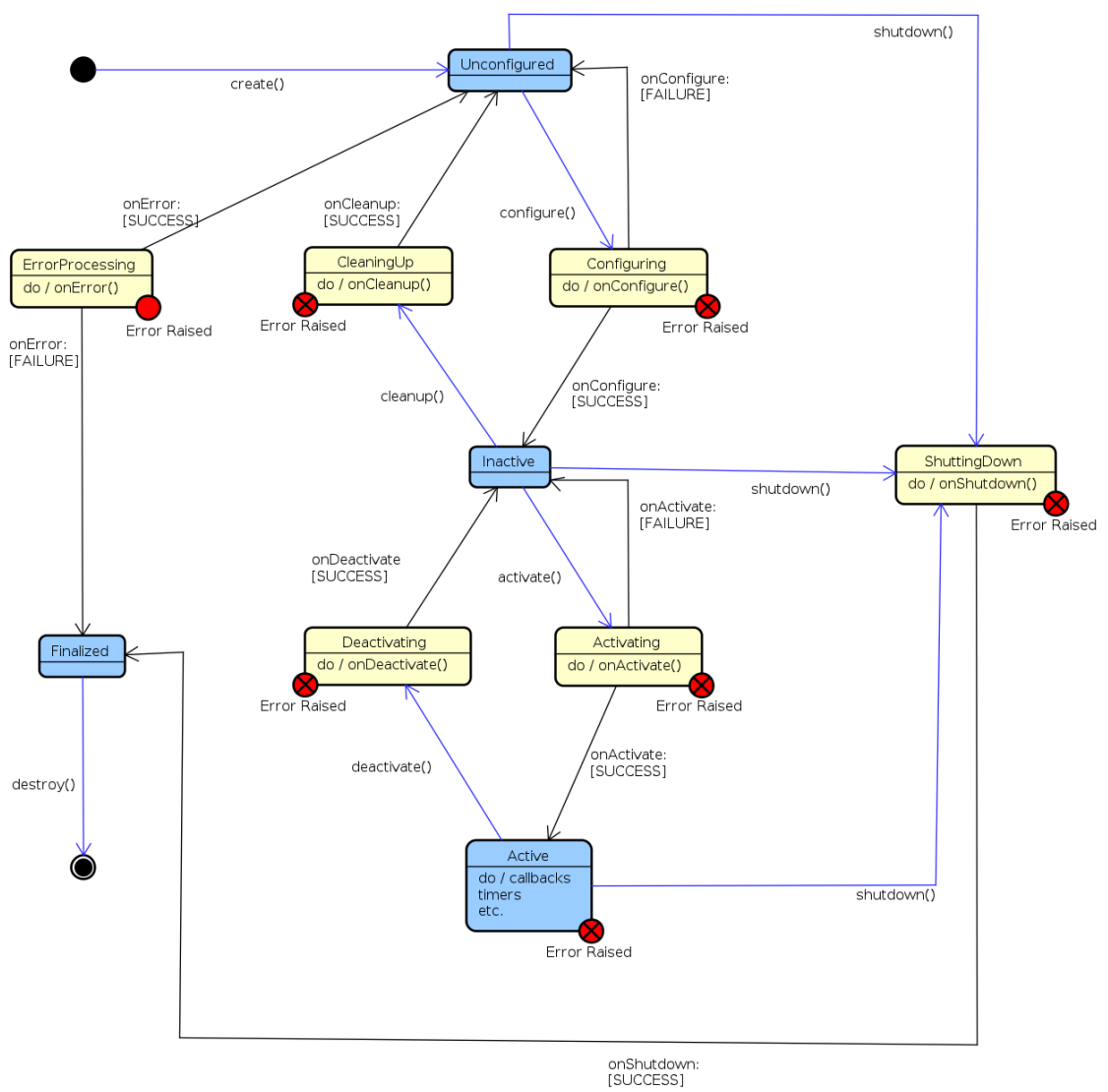


Figure 4.1: State Machine of LifecycleNode

Nodes can also be composed within the same process using the **Composition** feature, which improves performance by bypassing middleware overhead[7].

³<https://docs.ros.org/en/rolling/index.html>

- **Topics:** Topics are used for communication between nodes to share streams of messages in a publisher-subscriber structure.

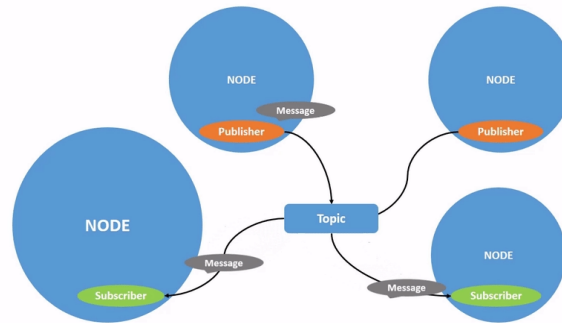


Figure 4.2: Topic process

- **Service:** Services are based on the request-response model. When a node with a service client sends a Request Message to a node with a service server, it receives a Response Message from that specific server.

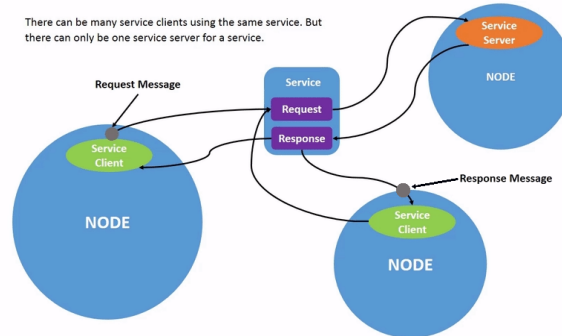


Figure 4.3: Service process

- **Parameters:** Parameters are configurations of a node that can be modified without changing the source code.
- **Actions:** Actions are designed to support long-running tasks that require feedback and the capability to interrupt when needed. They combine two services (goal and result) and a publisher-subscriber mechanism (feedback). A node sends a goal request and receives an acknowledgment, then starts publishing feedback messages while executing the task, and finally responds with the result.

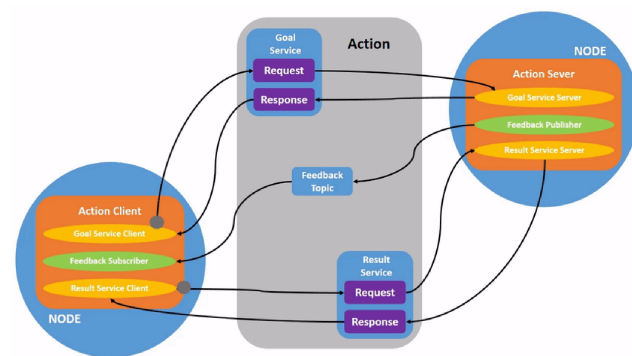


Figure 4.4: Action process

4.2 Tools of ROS

Now, I will describe the tools that are built into ROS for developing robotic systems:

- **Launch:** A file structure that allows the simultaneous start of multiple nodes with a set of parameters.
- **rqt:** A graphical user interface (GUI) tool for ROS. It includes a series of plugins that facilitate development, monitoring, and diagnostics of the ROS system.
- **rqt_console:** A plugin for rqt that visualizes the logs generated by the ROS. è solo un plugin del precedente
- **Rviz:** A 3D visualization tool for the Robot Operating System. It allows visualization of sensor data and state information from ROS.

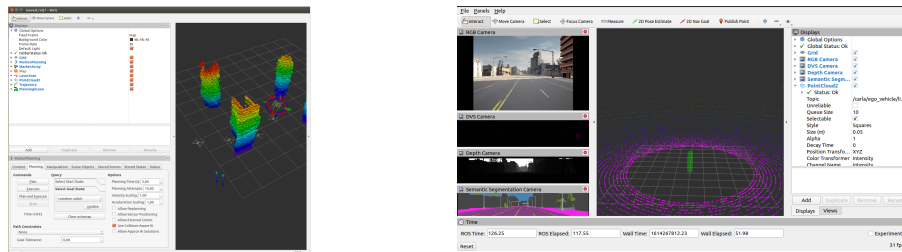


Figure 4.6: Examples of visualization of ROS data with Rviz

- **roscdep:** A command-line utility for installing system dependencies required by ROS packages.
- **sros2:** A security extension for ROS 2 that provides tools and instructions to apply security policies to DDS (Data Distribution Service).

Another important category of tools that are integrated with ROS is simulation environments. These environments allow developers to test their code on simulated robots under various conditions, which helps identify bugs that could potentially harm a real robot or its surroundings. Simulation tools provide realistic physical behavior for all components of a robotic system. According to the ROS documentation, two main simulation tools are commonly used: Gazebo and Webots. Among them, Gazebo is also maintained by Open Robotics and is widely adopted in many projects, including the one discussed in this paper.

Gazebo offer:

- **ROS Integration:** A bridge that converts Protobuf messages to ROS messages, enabling communication between Gazebo and ROS.
- **Performance tools:** Features for distributed simulation across servers, dynamic asset loading, and adjustable time steps to optimize simulation performance.

- **Realistic simulation:** Support for a variety of sensors, including their noise models, within a 3D world rendered using OGRE 2.1⁴ and powered by multiple physics engine, the default one is DART⁵.
- **Extensibility:** A modular design that supports plugins, allowing developers to run custom code that interacts directly with the simulation environment.

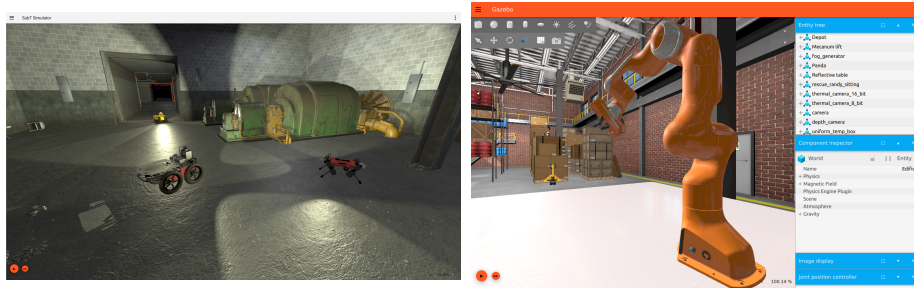


Figure 4.8: Examples of Gazebo

⁴<https://wiki.ogre3d.org/Home>

⁵<https://dartsim.github.io/>

Chapter 5

Introduction to NAV2 and Google Cartographer

Multiple tools are available to enable a robot to move autonomously within an environment. One of the most complete systems is Nav2 ¹ [10], the successor to the ROS 1 Navigation Stack.

5.1 Description of Nav2

Nav2 provides a comprehensive set of tools for perception, planning[13], control, localization, and visualization, allowing any robot to navigate autonomously and reliably, even in complex environments. It enables building a model of the environment from sensor data, dynamically calculating the optimal path to avoid obstacles, setting motor velocities, and executing navigation tasks using Behavior Trees.

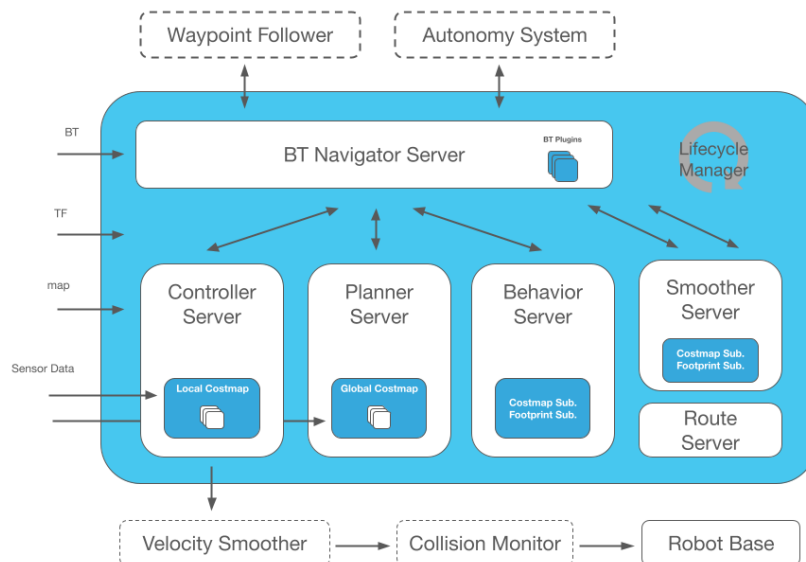


Figure 5.1: Nav2 architecture

¹<https://nav2.org/>

Nav2 introduces several improvements over ROS 1, including:

- **Behavior trees**[3]: Nav2 uses Behavior Trees to structure and execute tasks in a human-readable and modular way. These trees enable the creation of complex systems and can be tested with various tools. The BehaviorTree.CPP² library is used to implement them, and it allows linking different trees for flexible behavior management.
- **nav2_util LifecycleNode**: This is a wrapper around the standard `rclcpp_lifecycle::LifecycleNode` described earlier 4.1. It adds a bond connection to the Lifecycle Manager, which monitors the state of each node and can safely transition nodes through their lifecycle states in the event of a failure.
- **Action Server**: Described in 4.1, Nav2 uses action servers for Behavior Tree interactions, such as `NavigationToPose` (navigate to a specific location) and `NavigationThroughPoses` (navigate through multiple waypoints).

Nav2 is composed of several key nodes that work together to control the robot:

- **Controller Server**: Handles movement requests. It uses plugin-based modules such as a progress checker (to verify if the robot is making progress), goal checker (to determine if the robot has reached its destination), and a controller (to generate velocity commands using the **local costmap** while avoiding collisions).
- **Planner Server**: Computes the global path to the destination using sensor data and a plugin-based global planner loaded with the **global costmap**.
- **BT Navigator Server**: Implements the `NavigateToPose`, `NavigateThroughPoses`, and other task interfaces. It is a Behavior Tree-based implementation of navigation that is intended to allow for flexibility in the navigation task and provide a way to easily specify complex robot behaviors.

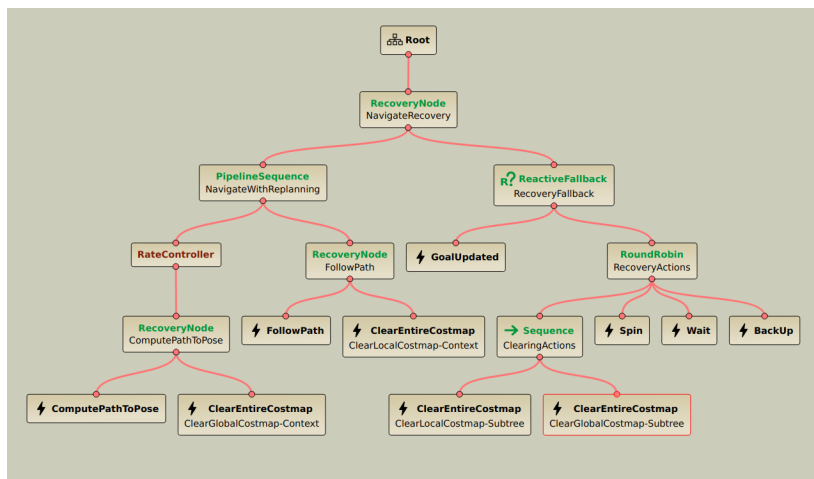


Figure 5.2: Humble Nav2 BT navigation `navigate_to_pose_w_replanning_and_recovery.xml` loaded with Groot2

²<https://www.behaviortree.dev/>

- **Smoother Server:** Smooth the path minimizing sharp turns and rapid rotations.
- **Behavior Server:** Handles specific behavior actions like spinning, backing up, and waiting.
- **Waypoint Follower:** An example of orchestrated control (an **Autonomy System**), this node takes a list of waypoints and uses the `NavigateToPose` action server to move between them, optionally performing custom tasks (e.g., taking a photo) at each waypoint through plugins.
- **Route Server:** Calculates routes through a predefined navigation graph instead of using free-space planning.
- **Velocity Smoother:** Smooths velocity, acceleration, and deadband signals sent to the robot to ensure safe and smooth motion.
- **Costmap** is a representation of the environment using a regular 2D grid of cells with a cost (unknown, free, occupied, or inflated) and is used for generate the global plan or to compute local control efforts.

The next things that Nav2 relies on for controlling the navigation are:

- **state estimation** of the robot that generate a suitable TF tree, a tree structure of coordinate frames generated using the transform library[4], based on the REP 105, that is should provide this structure: `map` $\rightarrow$ `odom` $\rightarrow$ `base_link`. According to the REP 105 they are defines as:
 - `map` : A world-fixed frame that may change in discrete jumps. It's useful for long-term global reference but not suitable for short-term localization.
 - `odom` : A continuous world-fixed frame that may drift over time. It provides short-term local reference.
 - `base_link`: A robot fix frame.

The `odom` $\rightarrow$ `base_link` link can be estimated using various sensors. For improved accuracy, we use the `robot_localization` package[12], which provides nonlinear state estimation using Kalman Filter fusing the robot's odometry and the IMU data.

The `map` $\rightarrow$ `odom` link can be generate with a global positioning system like Simultaneous Localization And Mapping (SLAM) method[2] or using the one provided by Nav2, Adaptive Monte-Carlo Localization (AMCL).

- **map** can either be preloaded or built dynamically using a SLAM method. One SLAM method systems is Google Cartographer, which will be described in the next section. It is also the SLAM method that was used to generate the map for navigation.

5.1.1 Cartographer

Cartographer[5] is a SLAM method that operates using two main subsystems: a **Local SLAM** (called **Frontend**) that build a series of locally consistent **submaps** from successive scans and from the pose extrapolator with the possibility that it can drift during acquisition and a **Global SLAM** (called **Backend**) that reduces drift by performing loop closure—detecting when the robot has returned to a previously visited location. It uses scan matching against submaps and applies global optimization to align all submaps consistently and build a globally accurate map. Cartographer is capable of achieving mapping accuracy of 5 cm, but it requires extensive parameter tuning to reach optimal performance.

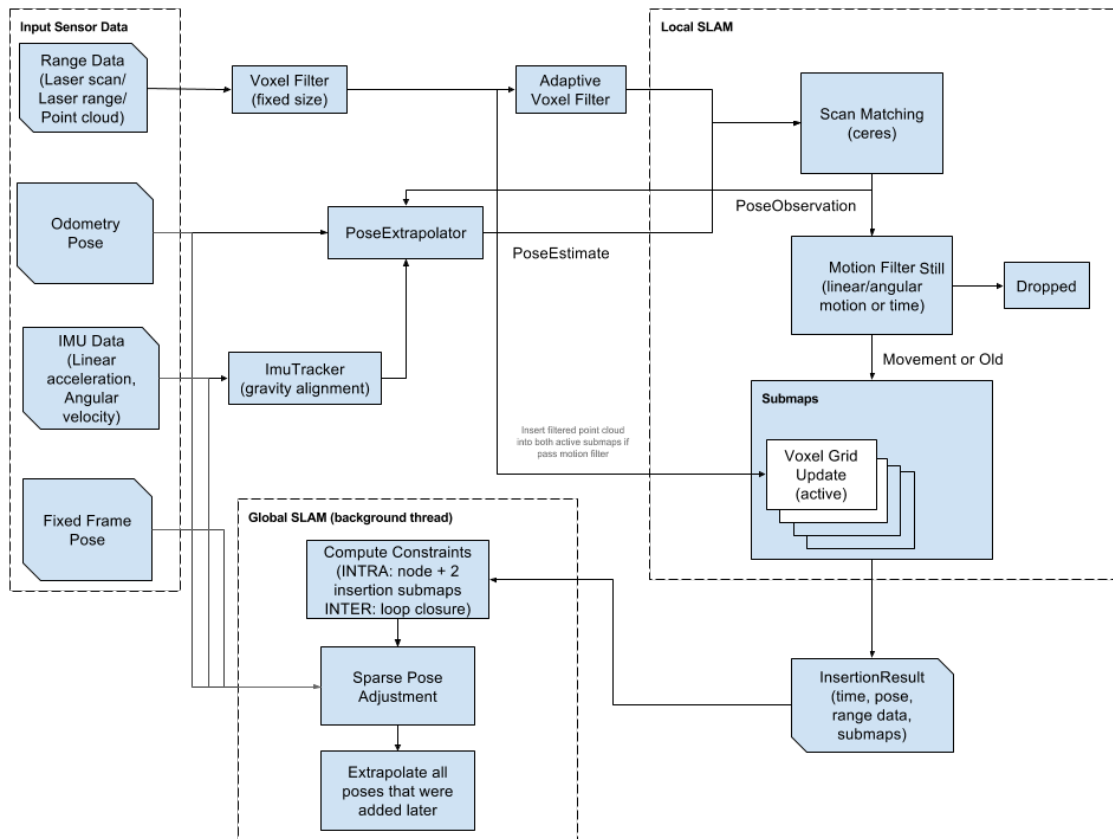


Figure 5.3: Cartographer Algorithm

Chapter 6

Review of related work on porting navigation stack from ROS 1 to ROS 2

The widespread adoption of the Robot Operating System (ROS) has introduced new use cases that were not originally considered during its development. Initially created as an academic project, ROS encountered limitations as users began applying it to more complex scenarios such as fleet management, deployment on embedded boards and microcontrollers, real-time systems, unreliable networks, and industrial applications. These emerging requirements highlighted architectural constraints in ROS 1, ultimately necessitating a complete redesign, which led to the development of ROS 2. Comprehensive documentation for ROS 2 is available at <https://design.ros2.org/>. Many companies have adopted the ROS 2 ecosystem, noting improvements in development workflows[9] and in specific subsystems like in Nav2[13]. New initiatives, such as micro-ROS¹, aim to bring ROS 2 capabilities to microcontrollers. Additionally, features like Node Composition offer improved performance optimization[7] while sros2 allow for a better security[15]. As outlined in these references, ROS 2 offers substantial advantages over ROS 1, making it the preferred framework for modern robotic applications. With this in mind, we will migrate the global planner using the official ROS 2 documentation² and various secondary sources³.

¹<https://micro.ros.org/>

²<https://docs.ros.org/en/rolling/How-To-Guides/Migrating-from-ROS1.html>

³https://industrial-training-master.readthedocs.io/en/melodic/_source/session7/ROS1-to-ROS2-porting.html

Chapter 7

Overview of open source fleet management systems, particularly Open-RMF

Currently, several fleet management systems are available:

- **Isaac Mission Dispatch**¹: A microservice-enabled fleet management system developed by NVIDIA that allows users to submit missions to multiple robots and monitor both robot and mission statuses. It provides connectivity between robots and the fleet management system but does not handle logistics such as task allocation or conflict resolution (e.g., intersecting robot paths). The implementation relies on the VDA5050 protocol, an industry standard for communication between cloud control services and mobile robots, and uses MQTT, a lightweight, publish-subscribe network protocol designed for devices with limited resources and bandwidth.
- **ROOSTER Fleet Management**²: An open-source ROS 1-based project that supports heterogeneous fleet management with task allocation, scheduling, and routing functionalities.
- **Open Robotics Middleware Framework (Open-RMF)**³: A free, open-source, and modular system that facilitates interoperability between multiple robot fleets and physical infrastructure such as doors, elevators, and building management systems.
- **Robofleet**⁴[14]: An open-source system based on ROS 1 that supports inter-robot communication, remote monitoring, and remote tasking for heterogeneous fleets of ROS-enabled service robots. It is specifically designed to handle network variability and to incorporate security control features.

Other frameworks provide the infrastructure needed to initiate fleet control:

- **Transitive Robotics**⁵: An open-source framework for full-stack robotics, designed to simplify the development of capabilities that connect robots, cloud

¹https://github.com/NVIDIA-ISAAC/isaac_mission_dispatch

²<https://github.com/ROOSTER-fleet-management>

³<https://www.open-rmf.org/>

⁴<https://github.com/ut-amrl/robofleet>

⁵<https://transitiverobotics.com/>

services, and web front-ends through a dynamically updating shared data model. It addresses common challenges in building full-stack robotic applications, such as unreliable networks, intermittently offline robots, and cross-device versioning.

- **Robot Web Tools**⁶: A collection of tools that enable communication with remote robots via web technologies, facilitating remote monitoring and interaction.

Given that Open-RMF offers control over multiple fleets, enables structured interaction between them to achieve optimal outcomes, provides a comprehensive toolset for system development, is based on ROS 2, and is open source, it will be used as the foundation for our fleet management architecture.

7.1 Open-RMF description

Open-RMF is a flexible and modular toolkit built on ROS 2 that facilitates interaction between heterogeneous robot fleets. It employs standardized protocols to enable communication with infrastructure, environments, and automation systems, optimizing the utilization of critical shared resources such as elevators, doors, narrow corridors, and other robots. By coordinating access to these resources, Open-RMF introduces system-level intelligence that helps prevent conflicts and ensures proper task and resource allocation. Figure 7.1 illustrates the architecture and operational workflow of Open-RMF.

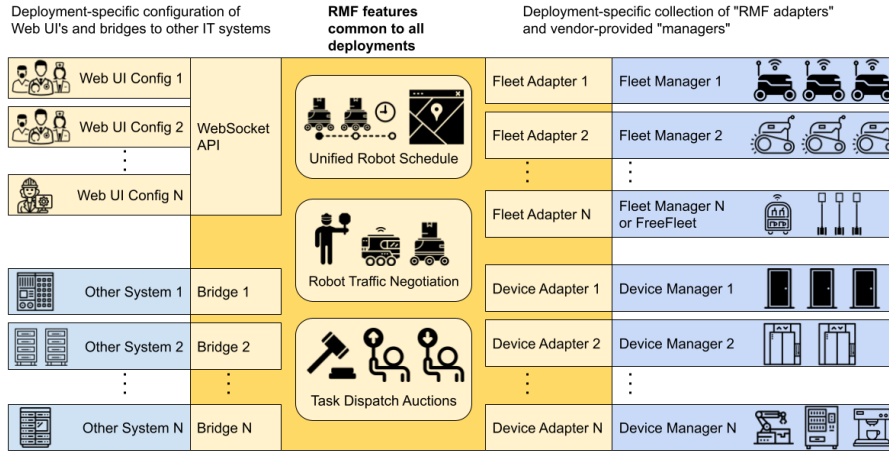


Figure 7.1: Open-RMF Structure

The fleets that can be controlled through Open-RMF can be categorized into three types:

- *Full Control*: Fleets of this type provide full status updates and allow complete control over path planning and execution.
- *Traffic Light*: These fleets also provide full status updates, but only allow limited control, such as pausing and resuming operations. Open-RMF has control over certain behaviors, but not over the specific paths.

⁶<https://robotwebtools.github.io/>

- *Read Only*: These fleets only provide status updates and cannot be controlled. Only one read-only fleet can exist in the system at any given time, as the presence of multiple read-only fleets would make it nearly impossible to avoid deadlocks or resource conflicts.

If a fleet lacks *No Interface*, it can lead to deadlocks in resource allocation and is incompatible with an RMF system.

The core of Open-RMF, known as `rmf_core`, is composed of several modular packages, each serving a distinct role within the system:

- `rmf_traffic`: Provides core scheduling and traffic management capabilities. It is based on a platform-agnostic **Traffic Schedule Database** and includes two main mechanisms for traffic deconfliction:

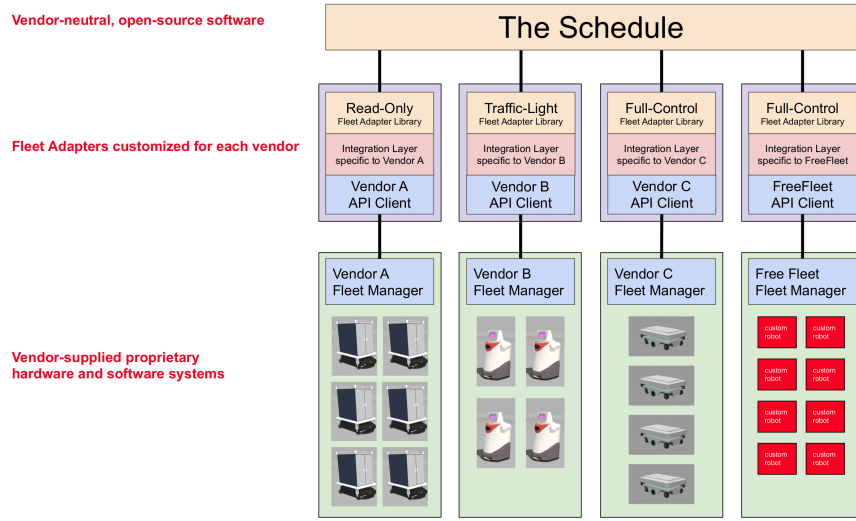


Figure 7.2: Open-RMF Schedule structure

- *Prevention*: When changes occur in the database (e.g., route cancellations), fleet managers can proactively replan routes to avoid conflicts.
- *Negotiation*: When a potential conflict is detected between two or more schedule participants, a conflict notice is issued to the relevant fleet managers. Each manager responds with a preferred path that accommodates other agents, and an external arbitration module selects the optimal path.

Conflict avoidance for Automated Guided Vehicles (AGVs) is implemented using a time-dependent extension to A* search. This algorithm considers both spatial and temporal dimensions, allowing it to generate conflict-free paths by accounting for the future movements of other agents. Support for Autonomous Mobile Robots (AMRs) is under active development.

- `rmf_traffic_ros2`: Provides the necessary ROS 2 interfaces for integrating `rmf_traffic` into ROS 2-based systems.
- `rmf_task`: Contains APIs and base classes for defining and managing tasks in RMF. The framework natively supports three types of task requests:

- *Clean*: For robots capable of cleaning floor spaces.
- *Delivery*: For robots tasked with transporting items between locations.
- *Loop*: For robots required to repeatedly navigate between two or more points.
- **rmf_dispatcher_node**: Contains the logic responsible for task allocation and management across multi-fleet systems. When a user submits a new task request, the dispatcher intelligently assigns it to the most suitable robot by querying each fleet adapter capable of performing the task. These adapters respond with bids indicating which robot is best suited, and RMF uses this bidding mechanism to determine the optimal assignment, as illustrated in Figure 7.3.

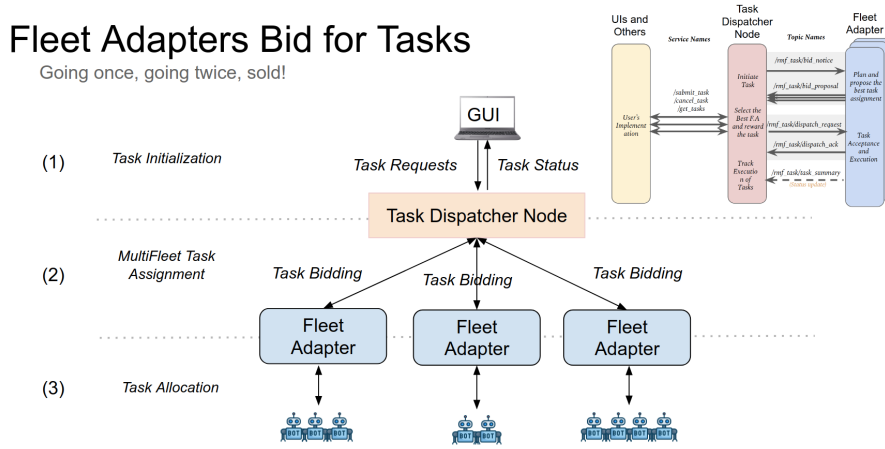


Figure 7.3: Open-RMF Bid Task

- **rmf_battery**: Provides battery consumption estimation models for different robot activities, aiding in energy-aware task planning.
- **rmf_ros2**: Supplies ROS 2 adapters, nodes, and Python bindings to integrate the RMF core components into ROS 2 applications.
- **rmf_utils**: A collection of utility functions and tools used across various RMF packages.

The following tools are available to support development with Open-RMF:

- **RMF demos**: A collection of demonstration packages showcasing the capabilities of Open-RMF. These serve as a useful starting point for development and experimentation.
- **Traffic Editor**: A graphical map editor used to define traffic patterns and interaction points (e.g., doors, lifts) that RMF uses for traffic management. It also supports the generation of Gazebo simulation worlds based on the designed maps.

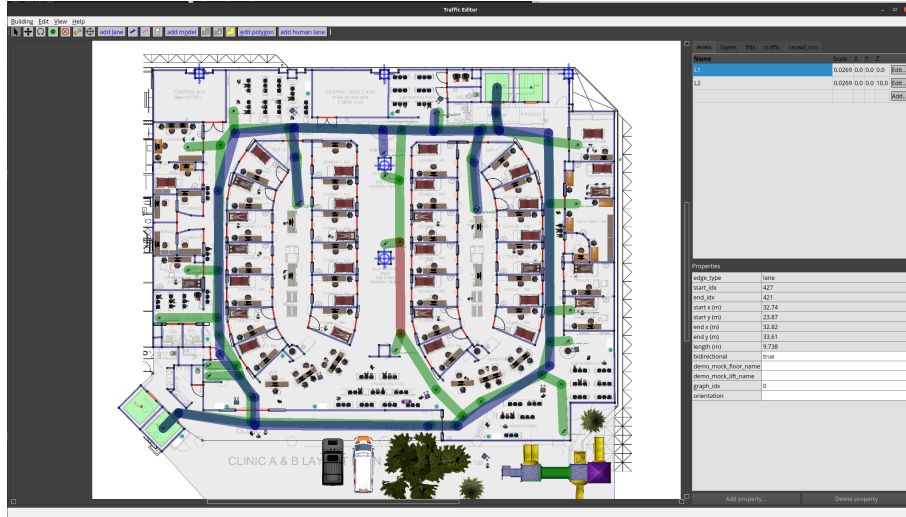


Figure 7.4: Traffic Editor

- **Free Fleet:** An open-source fleet adapter that enables the integration of standalone mobile robots into Open-RMF, allowing the creation of heterogeneous fleets.

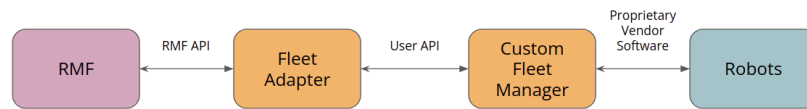


Figure 7.5: Fleet Adapter

- **RMF Schedule Visualizer:** An RViz plugin that provides visualization and monitoring of the RMF schedule, enabling users to observe and interact with traffic and task execution in real time.
- **RMF Web UI :** A web-based application for visualization and control of the Robotic Middleware Framework (rmf_core). It provides a user-friendly interface for monitoring system state and interacting with fleet operations.

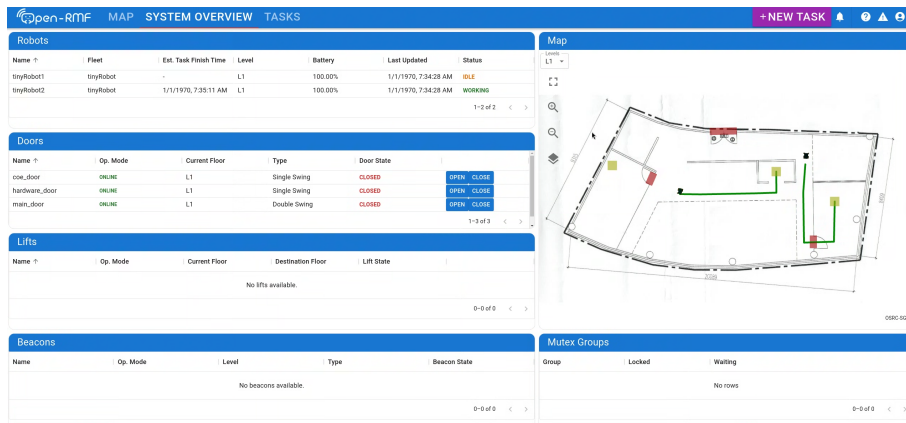


Figure 7.6: RMF Web

- **RMF Simulation:** A set of Gazebo plugins designed to simulate various aspects of building infrastructure—such as doors, lifts, and crowds—as well as robot behavior, including navigation and interactions with workcells.
- **Simulation assets:** Freely available 3D models and environment assets used to support simulation development and testing in Gazebo.

Part III

Methodology

In this part, we describe the entire procedure followed for building this project:

- Setup of our testing platform, which is based on the Rover Mini and a suite of onboard sensors.
- Setup the Nav2 navigation stack.
- The procedure used to port the Global planner from ROS 1 to ROS 2 and how it's insert in Nav2
- And the tools used to integrate it in the Open-RMF system.

All the source code developed for this project is publicly available in the following GitHub repository: [michbelle/myTesiCode](https://github.com/michbelle/myTesiCode)⁷. The repository also includes the necessary components for simulating the platform.

The simulation environment is containerized using Docker, which provides a consistent and OS-independent base system. The Docker image is built on top of `osrf/ros:jazzy-desktop-full` and includes all the packages developed for this thesis, as well as the required Open-RMF tools and demo setups.

Tested were done with a i5-7600K CPU and a GeForce GTX 1060 6GB on Ubuntu 22.04. The software showed noticeable lag on this configuration; therefore, we recommend using more recent and powerful hardware for optimal performance.

⁷<https://github.com/michbelle/myTesiCode>

Chapter 8

Description of research platform and hardware used

The robot used in this thesis is a four-wheel drive (WD) skid-steering vehicle (see Figure 2.1). It has a maximum speed of 12.87 km/h and a payload of 45,36 kg. Its operational time ranges from 60 to 90 minutes under active usage, and up to 6 hours in idle mode. The manufacturer provides open-source control and simulation code on GitHub¹, compatible with the Gazebo simulator. However, this code is officially supported only for ROS 2 Humble, so that is the ROS 2 distribution chosen for this thesis.

The robot includes a USB interface that enables connection to a PC or single-board computer for control purposes. In this work, a mini PC is used to control the robot and also used to manage the entire navigation stack based on Nav2. The Mini PC is a **Beelink SER3-E-16512EJ0W64PRO** with a Ryzen 7 3750H processor which is a quad-core 8-thread 2.3 GHz mobile processor boosting to 4.0 GHz with a Radeon RX Vega 10 Graphics. The mini PC draws power directly from the robot, which also supplies 19V output. **This will reduce the operation time.**



Figure 8.1: Mini PC used

To improve odometry estimation, an additional sensor is mounted: the **IMU Wit-motion (WIT) JY901B**. This is a 9-Axis Combined IMU/Magnetometer/Altimeter (measure linear accelerations, angular velocities, Euler angles, magnetic field, barometry, altitude). It communicates via TTL/UART protocol. Sensor data are

¹https://github.com/RoverRobotics/roverrobotics_ros2

acquired using the Witmotion Sensors UART Connection Library[17], with a USB-to-serial converter, and are published in the ROS 2 environment using a dedicated node `witmotion_ros`, available on GitHub at `ElettraSciComp/witmotion_IMU_ros`².

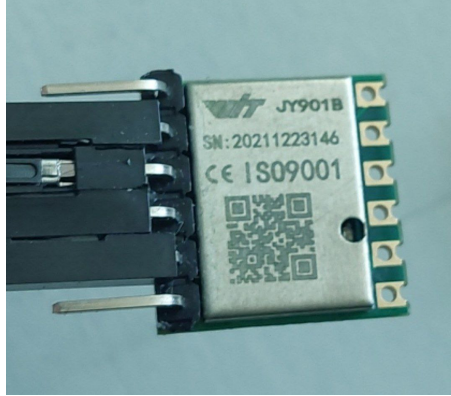


Figure 8.2: IMU used

For localization 5.1, a **2D laser scanner** is employed to provide environment perception and correct for odometry drift by maintaining a transformation link between the `map` and `odom` TF frames. The 2D laser scanner offers a cost-effective solution for environmental perception, providing dense spatial information that enables obstacle detection and basic environment recognition. Compared to alternative technologies [1][11], it requires significantly less computational power, making it a practical choice for lightweight and resource-constrained robotic platforms. The Laser scan used is **RPLIDAR S1** a 360 degree 2D laser scanner(LIDAR) with a distance range of 40m on white object and 10m on black ones, a sample rate of 9.2kHz, scan rate from 8 to 15 Hz, which mean an angular resolution of 0.313-0.587°, with an accuracy of $\pm 5\text{cm}$ and resolution of 3cm. The ROS 2 driver for the RPLIDAR S1 is available on Github at: `Slamtec/sllidar_ros2`³.



Figure 8.3: LIDAR used

²https://github.com/ElettraSciComp/witmotion_IMU_ros/tree/ros2

³https://github.com/Slamtec/sllidar_ros2

All components are physically mounted on the rover, using custom 3D-printed parts as shows in the following image:

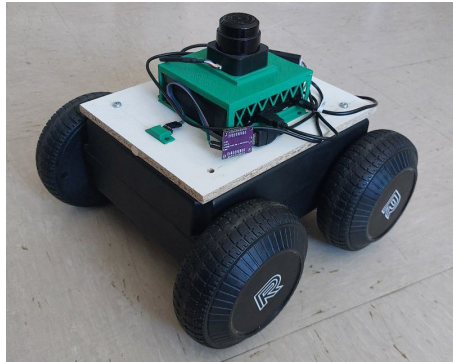


Figure 8.4: Robot used

To reflect this new hardware configuration, the original URDF (Unified Robot Description Format) file provided by the manufacturer was modified. This ensures accurate sensor placement relative to the `base_link` frame, which is essential for proper operation of sensor-dependent ROS 2 nodes.

A 3D model of the robot, including all coordinate frames (TF), is shown in this picture:

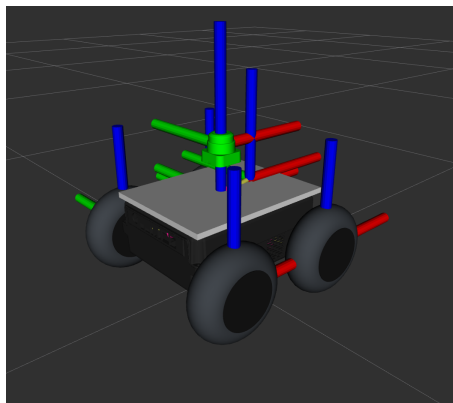


Figure 8.5: 3D model with TF

With the system fully assembled, the following ROS 2 launch files allow to get the data from the sensors connected on the robot's mini PC:

- `0.1r_sensorLaunch.launch.py`: Launches the nodes responsible for publishing data from the IMU and LIDAR sensors.
- `0.2r_mini.launch.py`: Publishes odometry data based on the built-in encoders of the robot.

While for simulation the following launch file can be used for starting the simulation:

- `0s_ign_mini_gazebo.launch.py`: Launch the simulation in gazebo of the robot with the sensors and using the node `ros_gz_bridge/parameter_bridge` share the data generate in Gazebo with ROS 2.

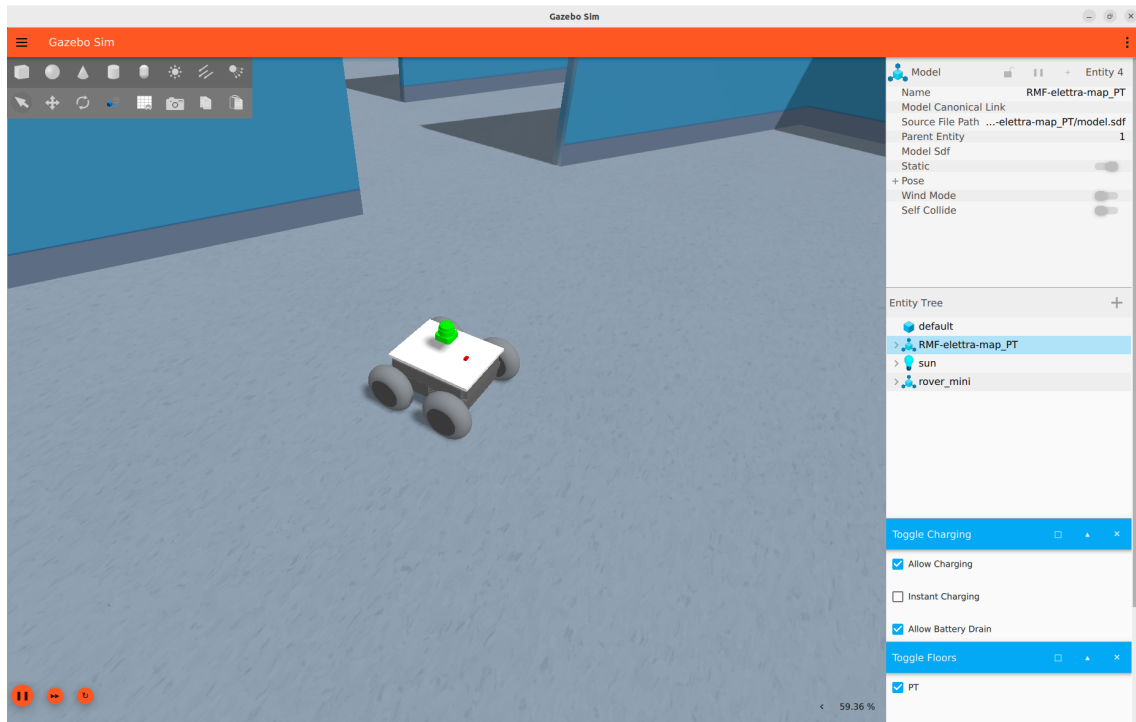


Figure 8.6: Model of the robot in Gazebo Ignition

Chapter 9

Steps taken to port the navigation stack from ROS 1 to ROS 2

In this chapter, we divide the system architecture according to the key components of the Nav2 stack that are utilized. Particular emphasis is placed on the process of porting the Global Planner module from ROS 1 to ROS 2, detailing the necessary adaptations and challenges encountered.

9.1 Odometry Estimation `odom` $\rightarrow$ `base_link`

As previously discussed, the raw odometry data provided by the rover are not sufficiently accurate, particularly in rotational movements[12]. This is a well-known issue in odometry estimation for skid-steering vehicles, which tend to produce significant drift and errors during turns.

To address this problem, we used the `robot_localization` package[12], specifically the `ekf_node`. This package enables the fusion of multiple sensor, even multiple instances of the same sensor type, and supports output in various ROS message formats, such as `nav_msgs/Odometry`. It also allows preprocessing of sensor messages prior to inserting into the estimation process. The filter performs continuous state estimation and is robust to intermittent sensor data loss, maintaining a coherent position estimate over time.

To launch the EKF node, the following launch file was wrote:

```
1<r/s>_robot_localizationEKF.launch.py
```

9.1.1 Configuration Odometry Estimation in `ekf_node`

As described previously and in the documentation of the node¹, we added the information about the sensors used to the configuration file. The principal information to include is the type of sensor and on which topic subscribe for the data, `odom0:odometry/wheels` and `imu0:imu/data`, and, for each sensor, we need to choose which components of the state measurement data we want to fuse using the following array, setting the corresponding values to `true` for those we wish to include and `false` for those we do not:

```
sensorX_config:[X,Y,Z,roll,pitch,yaw, $\dot{X}$ , $\dot{Y}$ , $\dot{Z}$ , $\dot{roll}$ , $\dot{pitch}$ , $\dot{yaw}$ , $\ddot{X}$ , $\ddot{Y}$ , $\ddot{Z}$ ]
```

¹https://docs.ros.org/en/noetic/api/robot_localization/html/index.html

In our case we used only the yaw velocity for the IMU while for the odometry we select only the velocity along the x and y.

It is also possible to tune the process noise covariance matrix Q , which represents the noise we add to the total error after each prediction step. Generally, the larger the value of Q relative to the variance of a given variable in an input message, the faster the filter will converge to the value indicated by the measurement.

9.2 Localization map $\rightarrow$ odom

In this section, we will describe how to acquire a map and use it to achieve the link between the `map` frame and `odom` frame using a localization method. This is necessary because, for long-term operation, the position of the robot in the world is not enough accurate and we need a method to compensate this error[2].

9.2.1 Mapping

We have already have a map, but for completeness, we provided the code for the following SLAM packages: `slam_toolbox`[6] (GitHub: SteveMacenski/slam_toolbox²) and `cartographer_ros`[5] (GitHub: ros2/cartographer_ros³). They can be launch using the following launch files:

```
X<r/s>_mappingW<slam/cartographer>.launch.py
```

To save the map, you can use the following command, which is valid for both `slam_toolbox` and `cartographer_ros`. This **command** will save the map in two files, a `.pgm` and a `.yaml`:

```
ros2 run nav2_map_server map_saver_cli -f map
```

When using Cartographer, it is also possible to save the map as a `.pdstream` using the **following command**:

```
ros2 service call /write_state cartographer_ros_msgs/srv/WriteState "{filename
ros2 interface show cartographer_ros_msgs/srv/FinishTrajectory.srv
ros2 service call /write_state cartographer_ros_msgs/srv/WriteState "{filename
```

²https://github.com/SteveMacenski/slam_toolbox/tree/ros2

³https://github.com/cartographer-project/cartographer_ros

9.2.2 Only localization

The map that was built using Cartographer is the following:

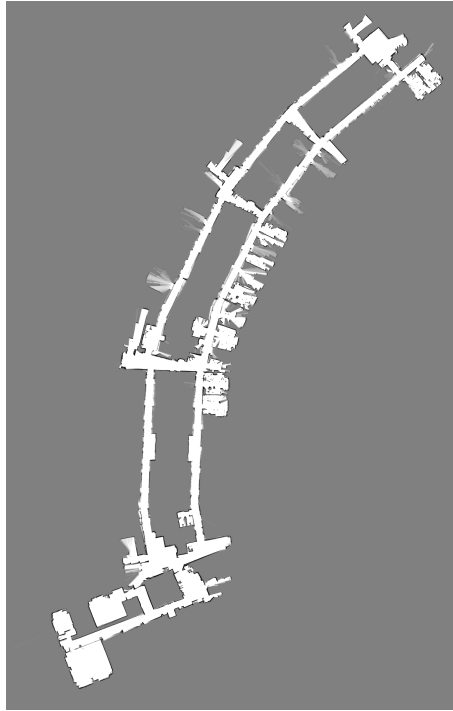


Figure 9.1: Map acquired with Cartographer

We attempted to use Cartographer solely for localization; however, due to the project's lack of maintenance, **we found it challenging to use it correctly**. As a solution, we opted for the method provided by the Nav2 package, specifically `nav2_amcl` and the `amcl` node.

But this has the problem that we need to convert the map acquired with cartographer as a `.pbstream` to a pair of `.pgm` and `.yaml` for the `amcl` node. This can be done using this **command**:

```
ros2 run cartographer_ros cartographer_pbstream_to_ros_map \  
  -pbstream_filename map.pbstream \  
  -map_filename map.pgm \  
  -resolution 0.05
```

Since certain walls of the Cartographer map were not solid black, the conversion brings some errors that lead to the misinterpretation of these areas as free zones and the planner returned a path that pass through them. **The solution of this problem was to add the walls manually using GIMP**, an image editor, on the .pgm file. We then obtained the following map:

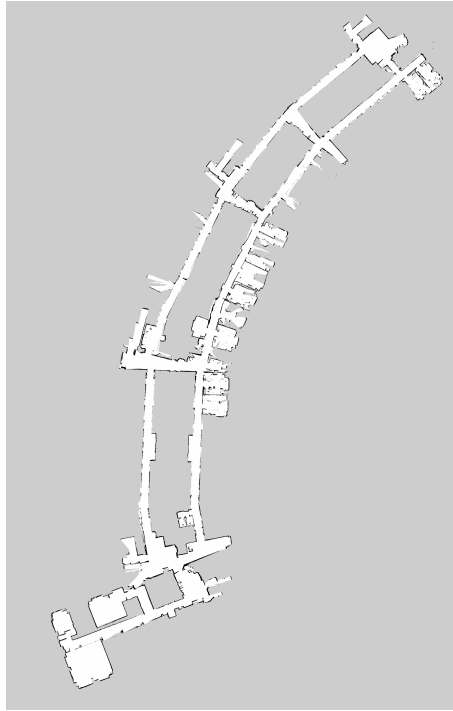


Figure 9.2: Map edited with Gimp and used

We have now all tools necessary for the localization and we have wrote the following launch file to start the localization:

```
3<r/s>_L<amcl/Cartographer>.launch.py
```

Now we have all the requirements to use the navigation stack Nav2.

9.3 Structure Navigation Stack Nav2

In this section we will discuss what components are used for the navigation stack and how we have ported the Global planner from ROS 1 to ROS 2.

9.3.1 Navigation Launch file

The launch file contains all the necessary processes required for the Navigation Nav2 system to operate. It implement also the composition structure (4.1) and the lifecycle manager (4.1). This is the list of all process used with a little description of what is used with them, **documentation is available at** ⁴ :

- `nav2_controller::ControllerServer` that use the following plugins:

⁴https://docs.nav2.org/setup_guides/algorithm/select_algorithm.html

- As `controller_plugins` use `nav2_mppi_controller::MPPIController` that is a Model Predictive Path Integral (MPPI) algorithm to track a path with adaptive collision avoidance.
- As `progress_checker_plugins` use `nav2_controller::SimpleProgressChecker` that checks whether the robot has made positional progress.
- As `goal_checker_plugins` use `nav2_controller::SimpleGoalChecker` that checks whether the robot has reached the goal pose.
- `nav2_smoother::SmootherServer` that use the following plugins:
 - As `smoother_plugins` use `nav2_smoother::SimpleSmoother` that take in an input path and smooth it using a simple and fast smoothing technique. It weights the initial path points and the smoothed path points to create a balanced result where the path retains its high level characteristics but reduces oscillations or jagged features.
- `nav2_planner::PlannerServer` that use the following plugins:
 - As `planner_plugins` use `dstar_global_planner/DStarGlobalPlanner` that is our global planner plugin and is a implementation of the D* informed incremental search algorithm with external optimizers[16].
- `nav2_behaviors::BehaviorServer` that use the following plugins:
 - As `behavior_plugins` use: `nav2_behaviors/Spin` performs an in-place rotation by a given angle, `nav2_behaviors/BackUp` which causes the robot to back up by a specific displacement, `nav2_behaviors/DriveOnHeading` that is which causes the robot to drive on the current heading by a specific displacement, `nav2_behaviors/Wait` that is , `nav2_behaviors/AssistedTeleop` that is allow user to teleop the robot avoiding commands that can create collision.
- `nav2_bt_navigator::BtNavigator` that use the following navigators:
 - As `smoother_plugins` use `nav2_bt_navigator::NavigateToPoseNavigator` and `nav2_bt_navigator::NavigateThroughPosesNavigator` that use the bt tree to navigate
- `nav2_waypoint_follower::WaypointFollower` that use the following plugins:
 - As `smoother_plugins` use `nav2_smoother::SimpleSmoother` that is
- `nav2_lifecycle_manager::LifecycleManager` that handling the lifecycle transition states for the stack in a deterministic way

The simulation configuration file contains minor differences due to the use of different versions of ROS 2: in simulation Jazzy while on robot Humble In the next section, , we will describe the key steps taken to port the global planner `DStarGlobalPlanner` from ROS 1 to ROS 2.

9.3.2 Porting ROS 1 DStarGlobalPlanner to ROS 2

As mentioned previously, the `DStarGlobalPlanner`⁵ is a implementation of the D* informed incremental search algorithm with external optimizers[16] for the ROS 1 `move_base` package⁶.

Since the code is written using ROS 1, it needs to be ported to ROS 2. The procedure involved splitting the process into two parts. In the first part we ported the base structure from ROS 1 to ROS 2 using the porting documentation⁷ and in the second part, we adjust the code to fit the structure of a planner plugin using the documentation from Nav2 : "Writing a New Planner Plugin"⁹. Is now described the 2 phases of porting the code, the first one is the Nav2 structure porting procedure and the second part adapt the old code to ROS 2.

Porting code from ROS 1 to ROS 2

In the `package.xml` file, which describe the package as an organizational unit for your ROS 2 code, the following changes were made:

- `buildtool_depend: catkin` → `ament_cmake`: `ament_cmake` is the new build system for CMake based packages. Many utilities for managing packages have been added compared to `catkin` and more information is available in the documentation¹⁰.
- `roscpp` → `rclcpp`: `rclcpp` is the new ROS client library for C++, built on `rcl`, which support the implementation of language specific ROS Client Libraries. The `rcl` communicates with `rmw`, the interface of middleware primitives that are used by the higher level ROS API's and handle the communication with the DDS.
- update the naming of all the packages.
- to inform `colcon` that the package uses `ament_cmake`, it is necessary to specify this in the `build_type`.

After that we edited the `CMakeLists.txt` and the major changes are:

- update the naming of all the packages.
- set the new standard of C++ to 17.
- `find_package()` required to be called for each package
- call `ament_export_dependencies()` to indicates which `ament` packages are necessary at runtime.
- declare the package in ROS using `ament_package()` at the end.

⁵https://github.com/ElettraSciComp/DStar-Trajectory-Planner/tree/ros2_port

⁶https://wiki.ros.org/move_base

⁷<https://docs.ros.org/en/rolling/How-To-Guides/Migrating-from-ROS1.html>

⁸https://industrial-training-master.readthedocs.io/en/melodic/_source/session7/ROS1-to-ROS2-porting.html

⁹https://docs.nav2.org/plugin_tutorials/docs/writing_new_nav2planner_plugin.html

¹⁰<https://docs.ros.org/en/rolling/Concepts/Advanced/About-Build-System.html>

Other minor changes are:

- Log calls are changed from `ROS_` to `RCLCPP_`
- Structure of some messages are changed (es. `geometry_msgs::PoseStamped` → `geometry_msgs::msg::PoseStamped`).
- The structure of the C++ program now required the use of smart pointer.

Other changes relate to ROS 2 are described in the next section because they are relate to Nav2.

Porting code from ROS 1 Navigation stack to Nav2

This is a list of the changes made for porting the planner to Nav2:

- Since it is a global planner, it inherits from new abstract interface for global planners, `nav2_core::GlobalPlanner`. As a result, we updated the `global_planner_plugin.xml` file changing the `base_class_type`.
- Because it is a plugin based on `nav2_core::GlobalPlanner`, it inherits 5 pure virtual methods that are necessary for correct operation: `configure()`, `activate()`, `deactivate()` and `cleanup()`, `createPlan()`. We will now describe each method:

- `configure()`: it's called when the node enters `on_configure` state, is used for perform declarations of ROS parameters and initialization of planner's member variables. The input parameters are:

- * `rclcpp_lifecycle::LifecycleNode::WeakPtr` A weak pointer to link to the base class.
- * The name of the planner.
- * A share pointer to the `tf` buffer
`std::shared_ptr<tf2_ros::Buffer>`.
- * A shared pointer to the costmap
`std::shared_ptr<nav2_costmap_2d::Costmap2DRos>`.

This replace the `initialize` method and the class initialization in ROS 1.

- `activate()`: Called when the node enters `on_activate` state, used for implement operations which are necessary before planner goes to an active state. In our case we allocate memory and fill our internal costmap `state_map`, which is used by the D* (`dstar`) algorithm.
- `deactivate()`: Called when the node enters `on_deactivate` state, used for implement operations which are necessary before planner goes to an inactive state. In our case, this method is not used and is left empty.
- `cleanup()` : Called when the node enters `on_cleanup` state, used for clean up resources which are created for the planner. In our case we destroys the allocated memory used by `state_map` and `dstar`.

- `createPlan()`: Called when planner server demands a global plan from a start and goal pose. It required the start and goal position as `geometry_msgs::msg::PoseStamped` and return a path `nav_msgs::msg::Path`. Replace the `makePlan` used in ROS 1.
- At the end of the C++ code we write this two rows that are used to export the planner:

```
#include "pluginlib/class_list_macros.hpp"
PLUGINLIB_EXPORT_CLASS(<type of the plugin class>, <type of the base class>);
```

- Because it's a plugin, we need to add a plugin declaration creating a file `global_planner_plugin.xml`, where is located the `package.xml`, that describe the plugin and edit the CMake file as described in the documentation for plugins¹¹.

Building the project

Finally, we build the system using the new build tool, `colcon`¹². It is a command line tool to improve the workflow of building, testing and using multiple software packages. It automates the process, handles the ordering and sets up the environment to use the packages.

After compiling the code we have the global planner ready to be used.

¹¹<https://docs.ros.org/en/rolling/Tutorials/Beginner-Client-Libraries/Pluginlib.html>

¹²<https://github.com/colcon>

Chapter 10

Hardware adaptation process from Rover Zero to Rover Mini

Rimuovo il capitolo?

Chapter 11

Integration of the navigation stack with Open-RMF

To integrate a fleet in Open-RMF, we first need information about the structure of a building, like automatic doors or lifts that can be controlled, and the **route maps** that robots can use. The **route maps** are used to predict the navigation paths that the robots will take allowing RMF to avoid conflicts, this procedure is called "traffic management". All this information has to be stored in a file that can be generated using the **traffic editor tool**^{1 2}, a tool provided by Open-RMF. It now follows a description of how it works and how we have used it.

11.1 Traffic Management - Route Maps Generation

We start a new project `*.building.yaml` and then add the map generated by Cartographer and the drawing of the facility's plans. We then defined the position of important points in the map and design the traffic flow for the floor following the best practice as described here³. Now we use the following components of the traffic editor to insert all the information:

- **vertex**: They are waypoints or the definition of wall, door or else. The properties of a vertex can be:
 - `is_holding_point`: the robot is allowed to wait at this waypoint for an indefinite period of time.
 - `is_parking_spot`: robot is required to park itself.
 - `is_passthrough_point`: waypoint which the robot shouldn't stop.
 - `is_charger`: charging station for a robot.
 - `is_cleaning_zone`: used for the **Clean** task, indicate if it is a clean zone that is where to start the cleaning.
 - `dock_name`: used when the robot is executing their custom docking sequence. **send a command to rmf to dock the robot**.
 - `pickup_dispenser`: dispenser workcell for **Delivery** Task.

¹https://github.com/open-rmf/rmf_traffic_editor

²<https://osrf.github.io/ros2multirobotbook/intro.html>

³https://osrf.github.io/ros2multirobotbook/integration_nav-maps-strategies.html

- `dropoff_ingestor`: ingestor workcell for Delivery Task.
- `spawn_robot_type`: used in simulation to spawn a robot model.
- `spawn_robot_name`: used in simulation to identify a robot spawned.
- `measurement` allow to add a measurement to set the scale of the drawing. In our case we used the `know` width of the hallway.
- `door`: is used both for interacting with the real world than for simulation. If a robot is going to pass through a lane where is a door, RMF will request to open the door and also will add the model to the simulation.
- `traffic lane`: connect waypoints and are the paths for a fleet to follow, they are divided by groups and multiple fleet can be assign on a group. RMF suppose that are straight lines for planning. The lanes can be unidirectional or bidirectional and have a specific "orientation" if required the robot to navigate forward or backward.
- `fiducials`: used for align multiple floor and used as a link to ROS 2 map, are visible through multiple floor and give an information about their position respect the map.
- `lift`: used for align multiple floor and used as a link to ROS 2 map, are visible through multiple floor and give an information about their position respect the map.
- `walls`: used in simulation, add the walls to the world that will generate.
- `floor`: used in simulation, add the ground plane for robots to walk on. It can be edited to add holes for the lifts.
- `environment assets`: used in simulation, add Gazebo models of different furniture.

These decision were taken place when planning for the route maps:

- In the hallways can pass only one robot.
- Holding points (leaf nodes) are added along the corridor or near the begin or the end of it for giving way to one another. For some task waypoint is suggested to add a very low limit speed to avoid that RMF will use them as holding points.
- Adding mutex lanes or waypoints that allow, at any point of time, only one robot to occupy any waypoint or lane belonging to the same mutex group. This allows to avoids deadlocks.

At the end we obtained the following map:

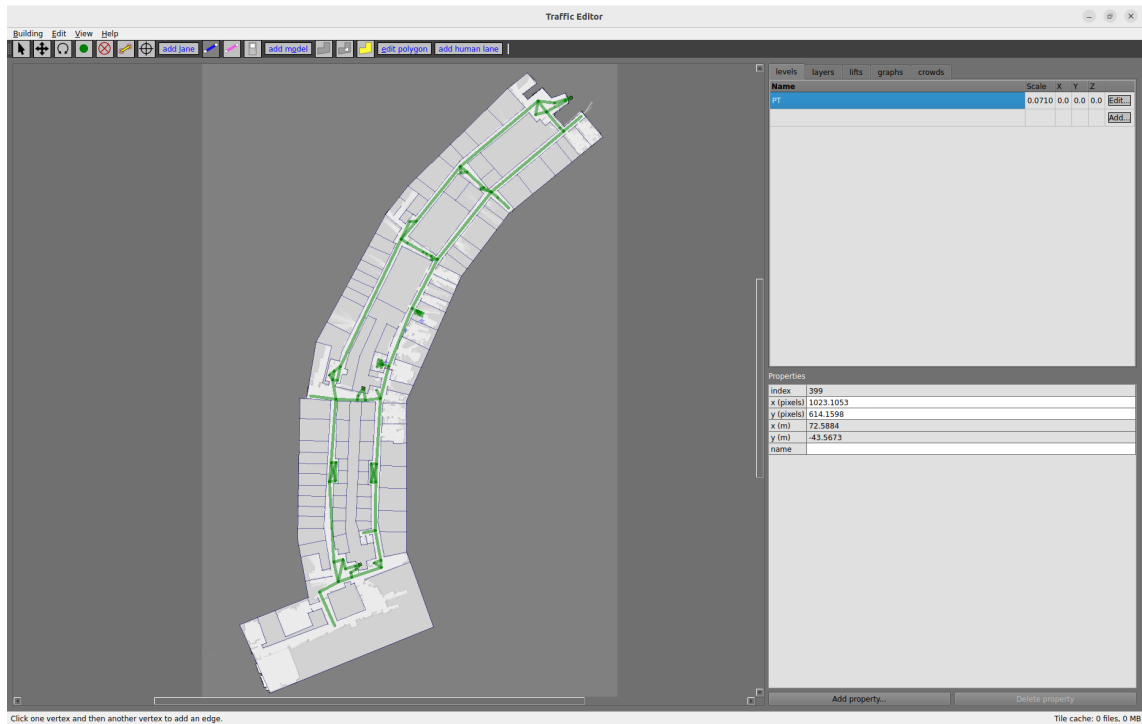


Figure 11.1: Traffic editor with Elettra's route map

Then it is possible to generate each route maps from each graphs *.yaml using the command:

```
ros2 run rmf_building_map_tools building_map_generator nav \
    ${building_map_path} ${output_nav_graphs_dir}
```

Then we can use the following command to generate the map that can be used in simulation:

```
ros2 run rmf_building_map_tools building_map_generator ignition \
    ${building_map_path} ${output_world_path} ${output_model_dir}
```

While to download the models used during the configuration of the app use:

```
ros2 run rmf_building_map_tools building_map_model_downloader \
    ${building_map_path} -f -e ~/.gazebo/models
```


For the simulation we obtained the following map:

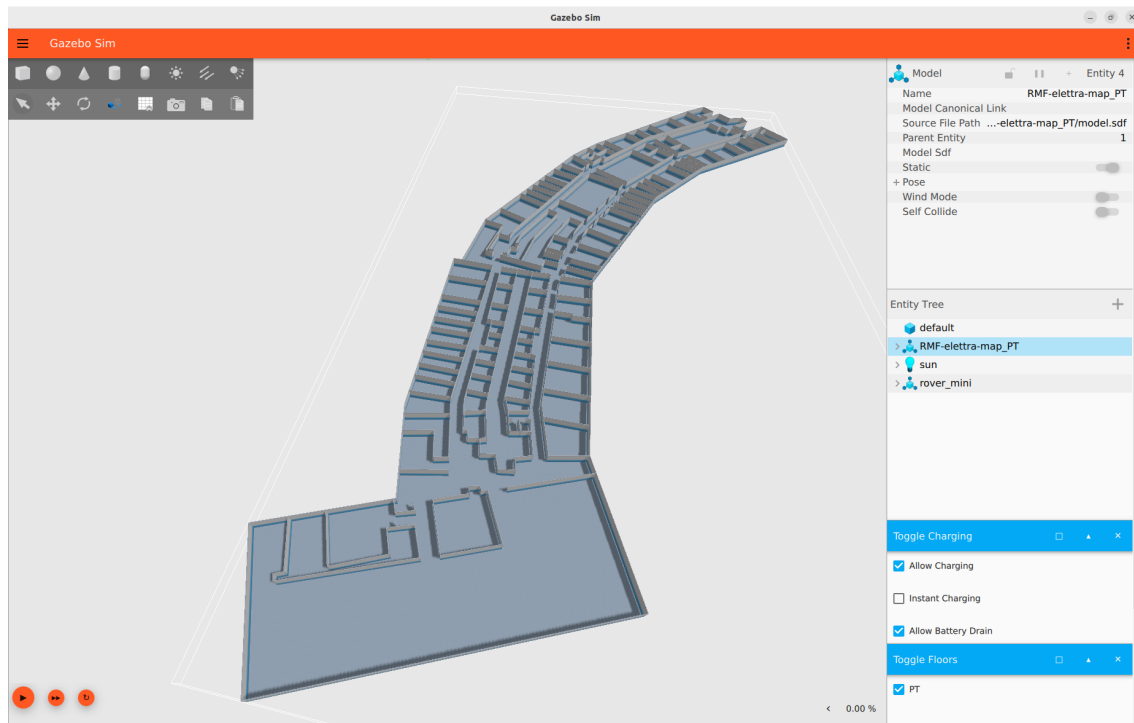


Figure 11.2: Map generated using traffic editor

Added all the information we can now launch the core of RMF.

11.2 RMF core

This is the launch file for starting the core of RMF:

`0rmf_mini_world_rmf_core.launch.xml`

And it launch the following nodes, that are the structure of the core of RMF:

- `rmf_traffic_schedule` : Traffic Schedule, **control the path structure and publish the itinerary topics, control the conflicts,**
- `rmf_traffic_blockade` : the traffic Blockade Moderator, **control checkpoint information**
- `building_map_server` : publish the data of the map like points of interes like parking spot
- `visualization.launch.xml`: optional node that allow to launch all the tool to see and control RMF using rviz2 **provare a rimuovere**
- `door_supervisor`: using ROS structure publish the request of control the door
- `lift_superviso`: using ROS structure publish the request of control the lift

- **rmf_task_dispatcher**: Dispatcher node will allow you to dispatch submitted task to the best fleet/robot within RMF.
- **queue_manager**: if request Enable the reservation node for parking spot de-confliction

Now that we have the core system launched we need to connect ours fleets using the fleet adapters. As said before Open-RMF gives a Free Fleet that is compatible with ROS 2 and Nav2. In the next section we will describe in detail how it work and how we have used it.

11.3 Free Fleet

Free fleet⁴ is a python implementation of the `full_control` Open-RMF Fleet Adapter⁵, that is the general template for communicate with a fleet manager, with the priority to control robot under ROS 1 and ROS 2. It uses **zenoh** as a communication layer between each robot and the fleet adapter. Following is view on how the free fleet adapter communicate with ROS:

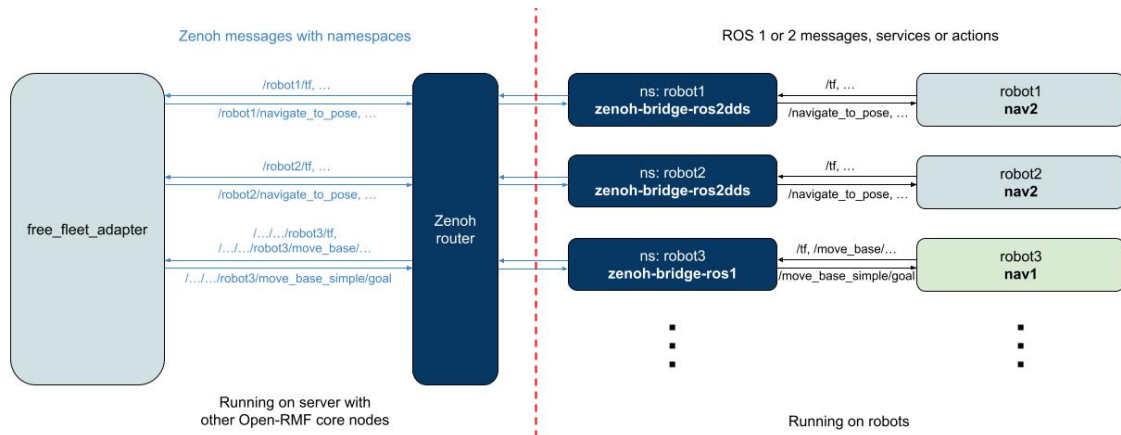


Figure 11.3: Free Fleet Adapter Communication

11.3.1 Setup fleet

For using the Free fleet we need to create the configuration file of the fleet. It a file that describe the robots of a fleet that have the same characteristics. These are the necessary information: name of the fleet, limits in velocity, their size and some mechanical info, some info their power management, what task they are capable of and a list of all robot that are inside this fleet with their name, what charger they use and the correlation in position between the map on the robot and the one used in RMF.

Setup the fleet, we need to setup Zenoh.

⁴https://github.com/open-rmf/free_fleet

⁵https://github.com/open-rmf/fleet_adapter_template

11.3.2 Setup Zenoh

Zenoh⁶ is a pub/sub/query protocol unifying data in motion, data at rest and computations. It blends traditional pub/sub with geo distributed storage, queries and computations, while retaining a well level of time and space efficiency. It allow to send only specific topic through the web reducing bandwidth to a specific server.

To setup this part of using Zenoh we need to: On a server is needed a Zenoh router that can be launched with (`$ zenohd`), but there can be multiple instance of Zenoh router for redundancy.

Then, as described in the documentation, on each robot and the server on witch run RMF there is the need to launch the standalone executable `zenoh-bridge-ros2dds` that bridges all ROS 2 communications using DDS over Zenoh. The configuration of this file depend on the machine. On the server it's configured to accept all the traffic `test se necessario avere il server` from each robot, while on the robot we need to setup:

- the `namespace` that is the same one associated to the file of configuration of the fleet
- enabling to publish the frame status of the robot (`./tf` and `./tf_static`) and the battery information (`./battery_state`)
- enabling the communication from the action servers that control the navigation(`./navigate_to_pose`)
- setup Zenoh specific configurations that define the mode of communication and the server's address, with its protocol and port that the client will `connect` and `listen` to (`tcp/<ip address server>:7447`)

11.3.3 Launching Free Fleet

After all of this we can use the following launch file to start the Free Fleet:

`1rmf_mini_fleet_adapter.launch.xml`

Its parameters are the fleet configuration file and the route maps that is going to use.

11.4 web interface

The it is possible to control the fleet remotly using a web interface. The open-RMF give 2 structure for remote control

- api server : allow execution
- dashboard : is the dashboard that connect to the api server and allow the control of the fleet

⁶<https://zenoh.io/>

Chapter 12

Testing and evaluation of the system

The testing have brigh up some problem in the design of the rover. If needed to move in office the robot with the laser sensor cannot see part of the obstacle, see image:



Figure 12.1: Cartographer Algorithm

can be increase the occupancy map bigger but the dimension of the legs chair wouldn't allow to pass through port. A solution is to move the laser to a lower position but this required a low profile laser, that need some filtering to remove the wheel and obtaining blind spot without adding the sensor on top or adding another sensor like a deep camera that give a 3D occupancy greed that can be processed to obtain a simple map.

The RMF commands from waypoint to waypoint: RMF commands the robot from

waypoint to waypoint, and it leaves the low level navigation to each robot's navigation stack. So if the robot chooses to use a non-straight route, it means that the RMF navigation graph might need to be re-drawn or reconsidered (if there are constant obstacles around), or the navigation stack of the robot should be tuned to allow tighter tolerances for static/dynamic obstacles. but how the robot chooses to navigate between them is out of RMF's control, since RMF does not have dynamic obstacle information, or if there are new static obstacles, it needs to be reflected into RMF's traffic navigation graph

Another thing that we discover both with the traffic editor and also using a better image manipulation program like gimp is that the image taken from the planimetry and the cartographer map are not well aligned. see image below:

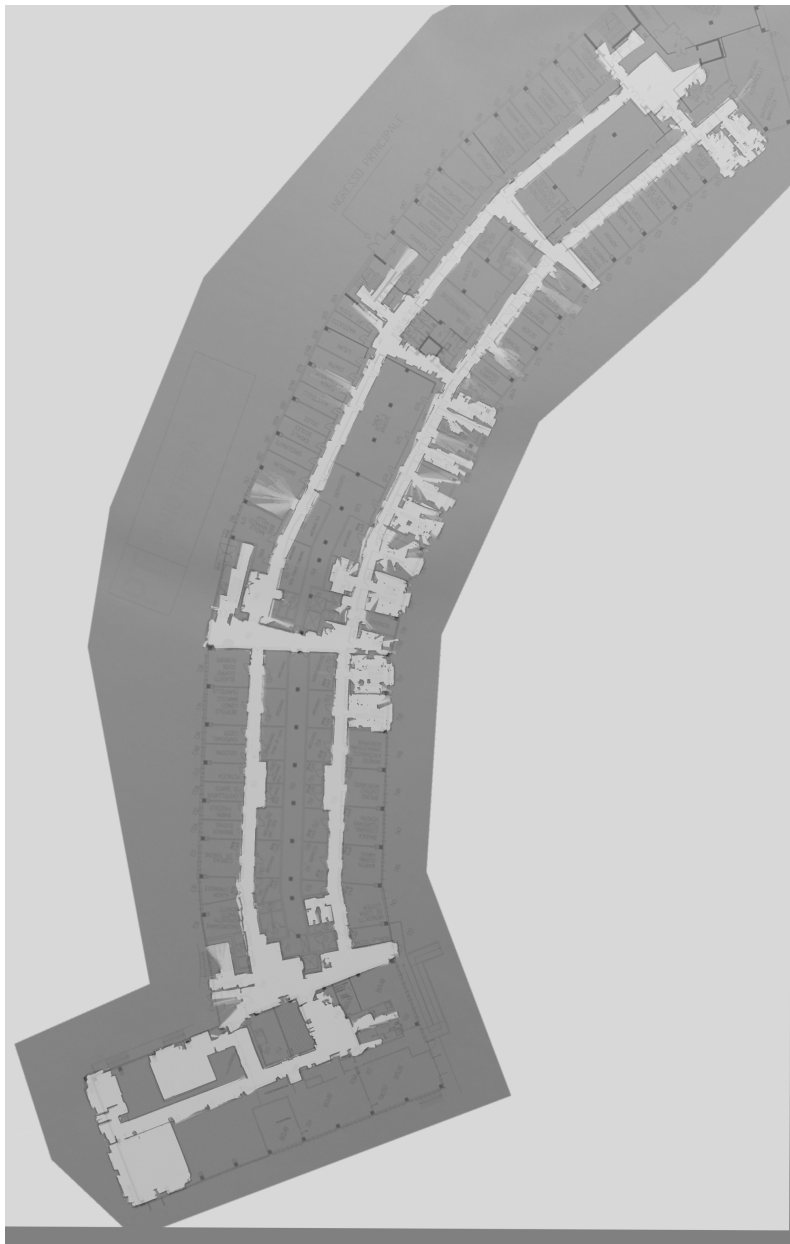


Figure 12.2: Overlap Cartographer Map and photo of floor plan

It's look like at each major rotation it overestimate the angle that the robot rotate when performing the mapping.

Part IV

Results and Analysis

Chapter 13

Presentation of results from testing and evaluation

Chapter 14

Comparison of system performance
before and after the porting process

Chapter 15

Discussion on the challenges encountered and how they were addressed

Part V

Conclusion and Future Work

Part VI

Appendices

Chapter 16

Simulation

In this part i will write some useful information for running the simulation. All test were done on debian based machine: To see the graphical interface launched in the docker container you need to have to share the access to the X server using the command `xhost local:root`.

today the tool for converting the data are <https://www.ros.org/blog/noetic-eol/>
implement also planning benchmark

describe and **use** composition in nav2 to obtain better performance [8]

Inoltre per gestire il traffico sulla rete utilizzeremo zenoh (Zero Overhead Network Protocol) ed il bridge per ros2 zenoh-bridge-ros2dds che è pub/sub/query protocol unifying data in motion, data at rest and computations. It elegantly blends traditional pub/sub with geo distributed storage, queries and computations, while retaining a level of time and space efficiency that is well beyond any of the mainstream stacks. data liberator protocol. Zenoh liberates data in several dimensions.

with chatgpt possiamo gestire il robot in due modi: si ferma ed esegue l'azione non comunicando con rmf (la linea rimane bloccata dal suo passaggio)

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